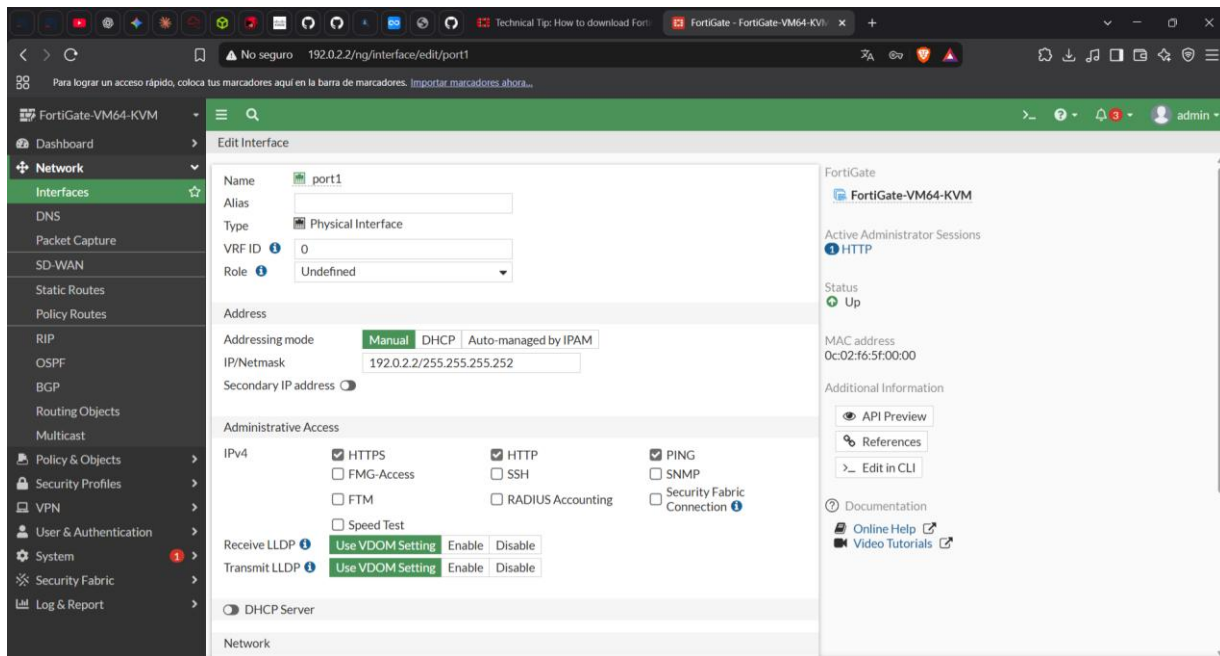


Evidencias de FortiGate

VPN Site-to-Site Cisco R2 ↔ FortiGate en GNS3

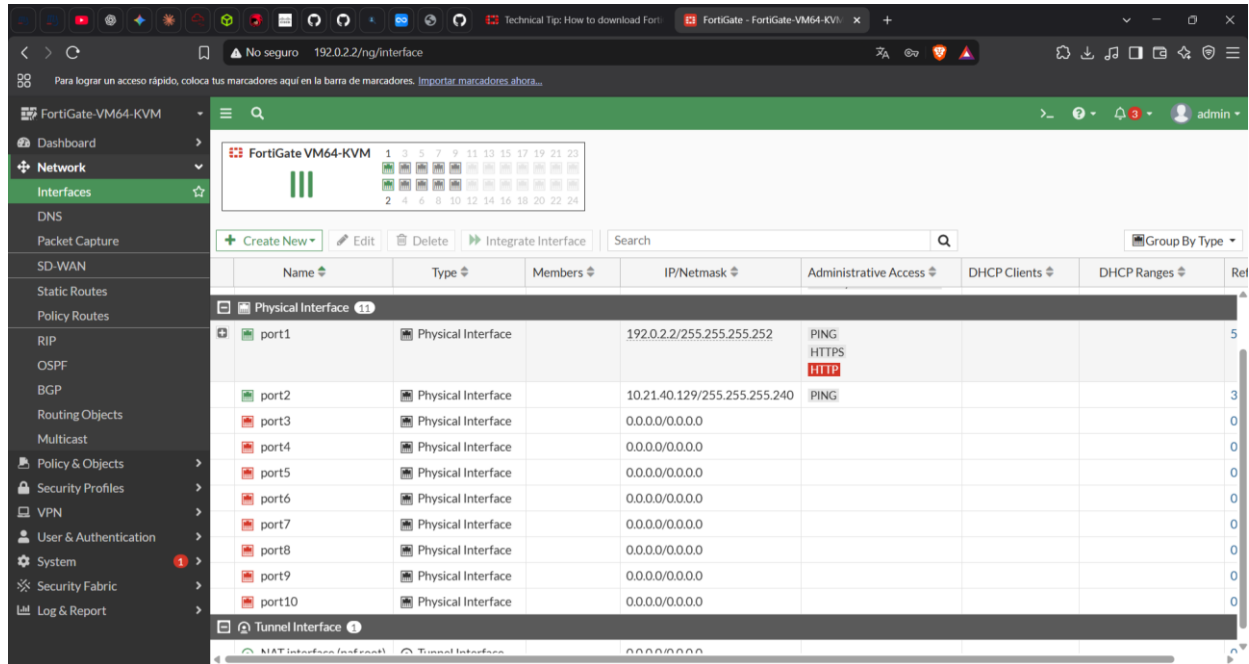
FortiGate - WAN (port1)

La interfaz WAN del FortiGate utiliza 192.0.2.2/30 y se conecta al router ISP mediante el segmento 192.0.2.0/30. En esta interfaz se habilitó acceso administrativo por PING, HTTPS y HTTP para permitir la administración desde la PC física.



FortiGate - Vista general de interfaces

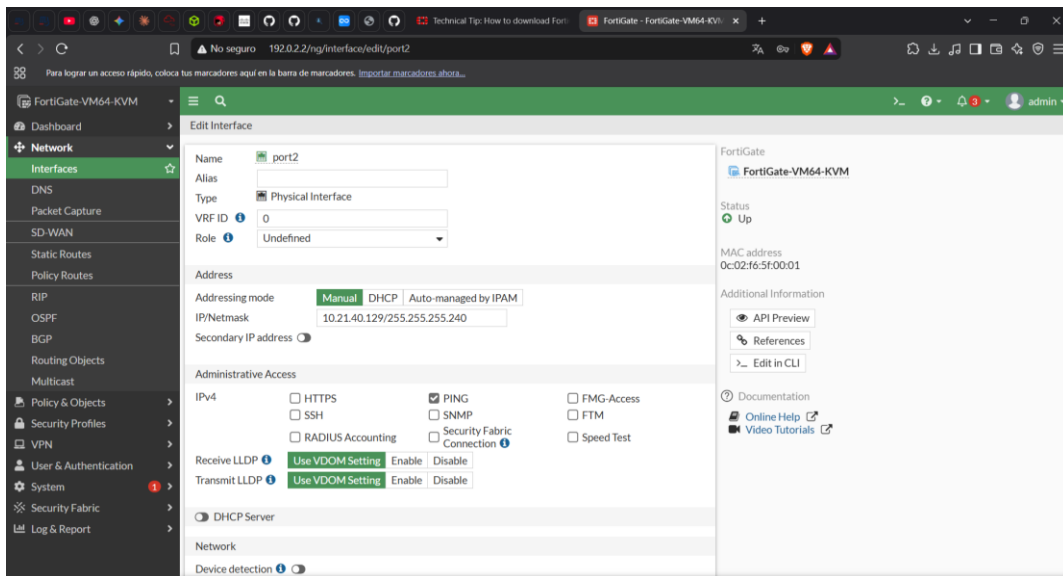
La vista general muestra las interfaces físicas utilizadas por el laboratorio y la interfaz lógica VPN-R2. port1 corresponde al enlace WAN, port2 conecta la red del servidor y VPN-R2 representa el túnel IPsec.



Name	Type	Members	IP/Netmask	Administrative Access	DHCP Clients	DHCP Ranges	Rel
Physical Interface 11							
port1	Physical Interface		192.0.2.2/255.255.255.252	PING HTTPS HTTP			5
port2	Physical Interface		10.21.40.129/255.255.255.240	PING			3
port3	Physical Interface		0.0.0.0/0.0.0.0				0
port4	Physical Interface		0.0.0.0/0.0.0.0				0
port5	Physical Interface		0.0.0.0/0.0.0.0				0
port6	Physical Interface		0.0.0.0/0.0.0.0				0
port7	Physical Interface		0.0.0.0/0.0.0.0				0
port8	Physical Interface		0.0.0.0/0.0.0.0				0
port9	Physical Interface		0.0.0.0/0.0.0.0				0
port10	Physical Interface		0.0.0.0/0.0.0.0				0
Tunnel Interface 1							
VPN-R2	Tunnel Interface		0.0.0.0/0.0.0.0				0

FortiGate - Red del servidor (port2)

La interfaz port2 utiliza 10.21.40.129/28 y funciona como gateway de la red del servidor. El servidor Ubuntu utiliza 10.21.40.130/28 con gateway 10.21.40.129.



Name	Alias	Type	VRF ID	Role
port2		Physical Interface	0	Undefined

Address
Addressing mode: Manual DHCP Auto-managed by IPAM
IP/Netmask: 10.21.40.129/255.255.255.240
Secondary IP address: [Off]

Administrative Access
IPv4: [Off] HTTPS [Off] SSH [Off] PING [On] SNMP [Off] FMG-Access [Off] FTN [Off] RADIUS Accounting [Off] Security Fabric Connection [Off] Speed Test [Off]
Receive LLDP: [On] Use VDOM Setting: [On] Enable [Off] Disable [On]
Transmit LLDP: [On] Use VDOM Setting: [On] Enable [Off] Disable [On]
DHCP Server: [Off]
Network: [Off]
Device detection: [On]

FortiGate - Rutas estáticas

Se configuró la ruta por defecto 0.0.0.0/0 vía 192.0.2.1, una ruta hacia la red física 192.168.18.0/24 para la administración desde la PC y una ruta hacia 10.21.40.0/25 mediante VPN-R2.

FortiGate-VM64-KVM

Dashboard

Network

- Interfaces
- DNS
- Packet Capture
- SD-WAN
- Static Routes
- Policy Routes
- RIP
- OSPF
- BGP
- Routing Objects
- Multicast

Policy & Objects

Security Profiles

VPN

User & Authentication

System

Security Fabric

Log & Report

192.0.2.2/ng/routing/static

Para lograr un acceso rápido, coloca tus marcadores aquí en la barra de marcadores. [Importar marcadores ahora...](#)

Create New

Edit

Clone

Delete

Search

Destination	Gateway IP	Interface	Status	Comments
0.0.0.0/0	192.0.2.1	port1	Enabled	
10.21.40.0/25	198.51.100.2	VPN-R2	Enabled	
192.168.18.0/24	192.0.2.1	port1	Enabled	

FortiGate - Políticas de firewall y NAT

Las políticas VPN-R2-to-SERVER y SERVER-to-VPN-R2 permiten tráfico entre RED-USUARIOS y RED-SERVIDOR con NAT deshabilitado. La política SERVER-to-INTERNET utiliza NAT para la salida del servidor hacia el ISP.

The screenshot displays the FortiGate VM64-KVM web interface. The left sidebar shows the navigation menu with 'Policy & Objects' expanded and 'Firewall Policy' selected. The main content area shows a table of firewall policies. The 'SERVER-to-INTERNET' policy is highlighted, showing details such as Source (RED-SERVIDOR), Destination (all), Schedule (always), Service (ALL), Action (ACCEPT), and Security Profiles (no-inspection, UTM). The 'VPN-R2-to-SERVER' policy is also visible, showing a warning icon and details like Source (RED-USUARIOS), Destination (RED-SERVIDOR), Schedule (always), Service (ALL), Action (ACCEPT), and Security Profiles (no-inspection, UTM).

FortiGate - Objetos de red

Se crearon los objetos RED-USUARIOS (10.21.40.0/25) y RED-SERVIDOR (10.21.40.128/28). Estos objetos se utilizan en las políticas de firewall y representan las redes protegidas por la VPN.

FortiGate-VM64-KVM

Dashboard

Network

Policy & Objects

Firewall Policy

IPv4 DoS Policy

Addresses

Internet Service

Database

Services

Schedules

Virtual IPs

IP Pools

Protocol Options

Traffic Shaping

Security Profiles

VPN

User & Authentication

System

Security Fabric

Log & Report

192.0.2.2/ng/firewall/address

Para lograr un acceso rápido, coloca tus marcadores aquí en la barra de marcadores. Importar marcadores ahora...

FortiGate - FortiGate-VM64-KVM

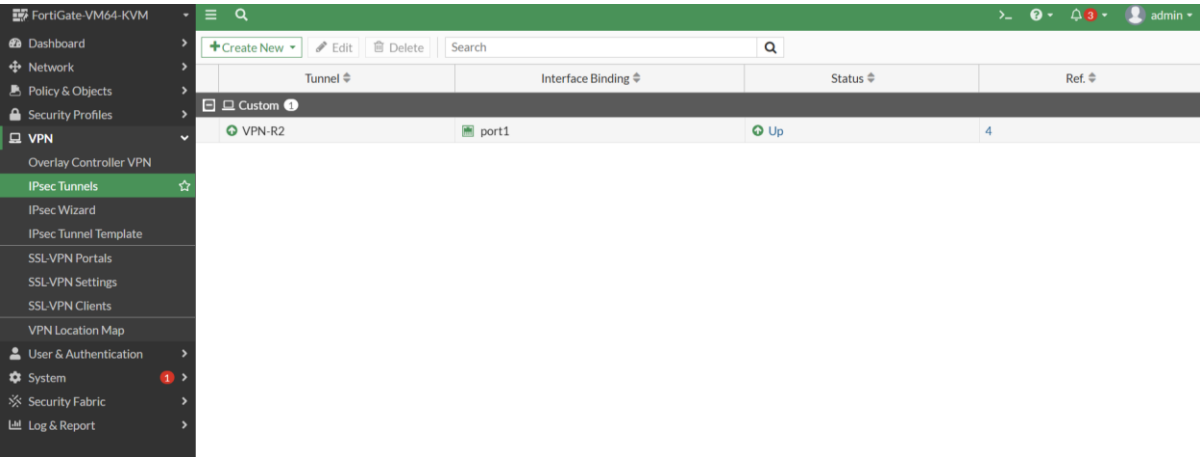
admin

Create New Edit Clone Delete Search

Name	Details	Interface	Type	Ref.
IP Range/Subnet				
FABRIC_DEVICE	0.0.0.0/0		Address	0
FIREWALL_AUTH_PORTAL_ADDRESS	0.0.0.0/0		Address	0
RED-SERVIDOR	10.21.40.128/28		Address	3
RED-USUARIOS	10.21.40.0/25		Address	2
SSLVPN_TUNNEL_ADDR1	10.212.134.200 - 10.212.134.210		Address	1
all	0.0.0.0/0		Address	1
none	0.0.0.0/32		Address	0
FQDN				
gmail.com	gmail.com		Address	1
login.microsoft.com	login.microsoft.com		Address	1
login.microsoftonline.com	login.microsoftonline.com		Address	1
login.windows.net	login.windows.net		Address	1
wildcard.dropbox.com	*dropbox.com		Address	0
wildcard.google.com	*google.com		Address	1
Address Group				
G Suite	gmail.com wildcard.google.com		Address Group	0
Microsoft Office 365	login.microsoftonline.com		Address Group	0

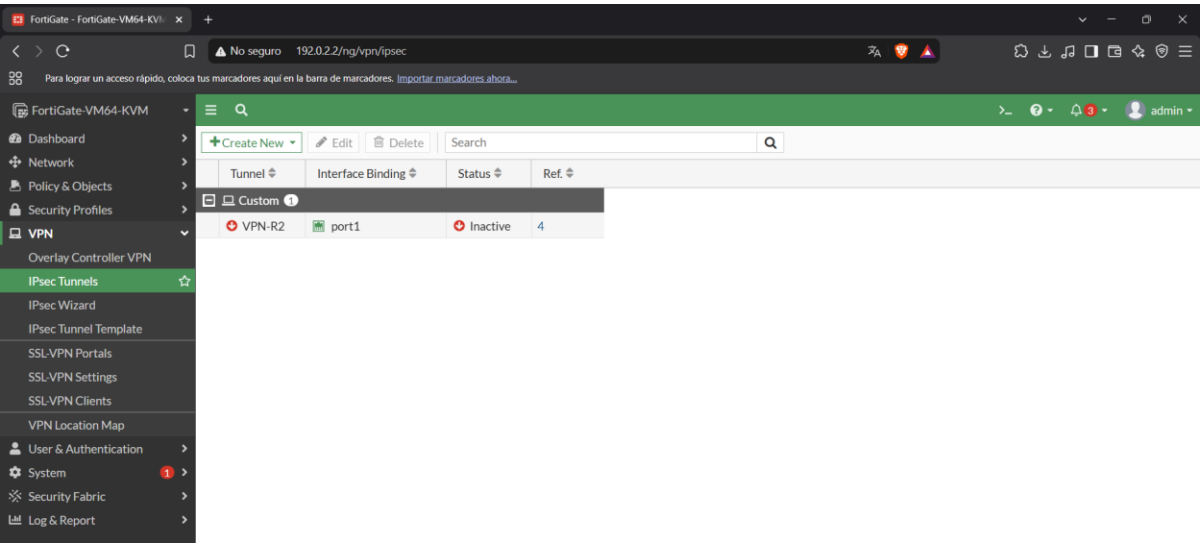
FortiGate - Estado del túnel IPsec

El túnel VPN-R2 aparece en estado Up sobre la interfaz WAN port1. Este estado confirma que el túnel IPsec se encuentra establecido entre el FortiGate y el router Cisco R2.



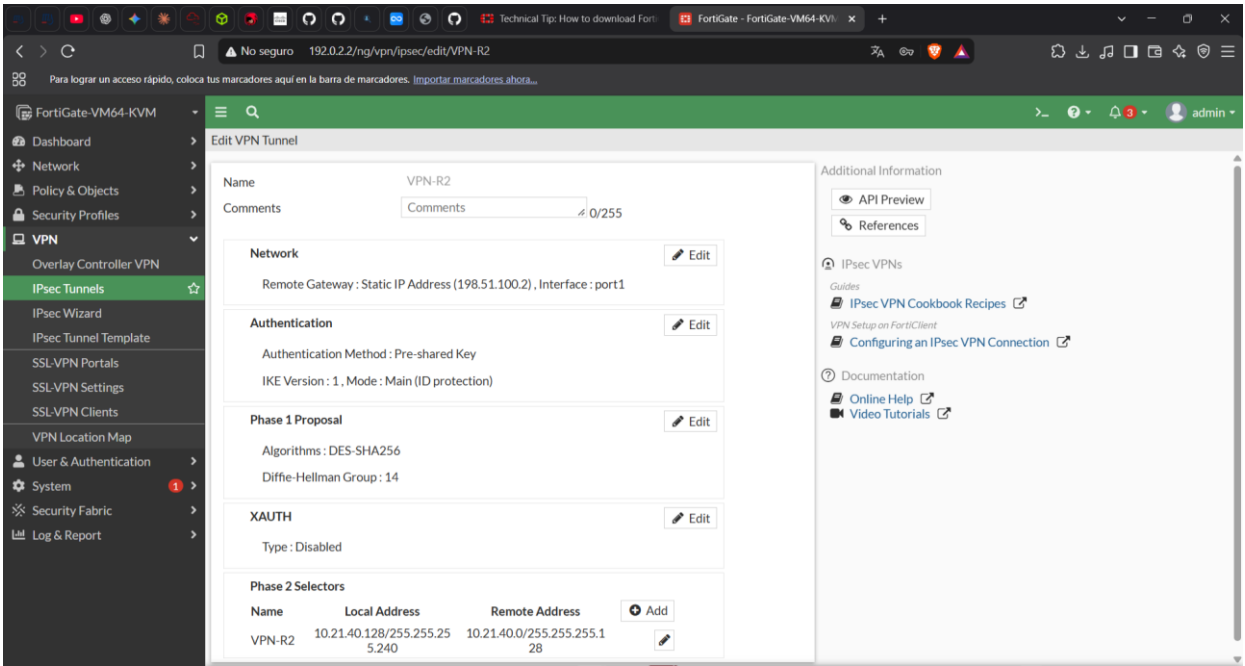
FortiGate - Túnel IPsec inactivo

Para validar que la comunicación depende de la VPN, se deshabilitó temporalmente VPN-R2. El túnel aparece Inactive y, en este estado, el acceso desde Windows al servidor deja de funcionar.



FortiGate - Detalle del túnel VPN

El túnel VPN-R2 utiliza como peer remoto 198.51.100.2, la interfaz de salida port1 y autenticación mediante Pre-shared Key. La clave compartida se mantiene oculta en la documentación pública.



FortiGate - Selectores de Phase 2

Los selectores de Phase 2 corresponden a la red local 10.21.40.128/28 y la red remota 10.21.40.0/25. Estas son las redes cuyo tráfico debe protegerse mediante IPsec.

FortiGate-VM64-KVM

Dashboard

Network

Policy & Objects

Security Profiles

VPN

Overlay Controller VPN

IPsec Tunnels

IPsec Wizard

IPsec Tunnel Template

SSL-VPN Portals

SSL-VPN Settings

SSL-VPN Clients

VPN Location Map

User & Authentication

System

Security Fabric

Log & Report

Edit VPN Tunnel

Authentication

Phase 1 Proposal

XAUTH

Phase 2 Selectors

VPN-R2

Authentication Method : Pre-shared Key

IKE Version : 1, Mode : Main (ID protection)

Algorithms : DES-SHA256

Diffie-Hellman Group : 14

Type : Disabled

Name	Local Address	Remote Address
VPN-R2	10.21.40.128/255.255.255.240	10.21.40.0/255.255.255.128

Edit Phase 2

Name

VPN-R2

Comments

Comments

Local Address

Subnet

10.21.40.128/255.255.2

Remote Address

Subnet

10.21.40.0/255.255.255

Advanced...

Additional Information

API Preview

References

IPsec VPNs

Guides

IPsec VPN Cookbook Recipes

VPN Setup on FortiClient

Configuring an IPsec VPN Connection

Documentation

Online Help

Video Tutorials

FortiGate - Phase 1 Proposal

La Phase 1 utiliza IKEv1 en modo Main (ID protection), cifrado DES, autenticación SHA256, Diffie-Hellman Group 14 y un Key Lifetime de 86400 segundos.

The screenshot displays the FortiGate web interface for configuring an IPsec Tunnel. The left sidebar shows the navigation menu with 'VPN' expanded and 'IPsec Tunnels' selected. The main content area is titled 'Edit VPN Tunnel' and shows the 'Authentication' section with 'Pre-shared Key' and 'IKE Version: 1, Mode: Main (ID protection)'. Below this is the 'Phase 1 Proposal' section, which is highlighted in yellow. It includes a table for Diffie-Hellman Group selection, where group 14 is selected, and a 'Key Lifetime (seconds)' field set to 86400.

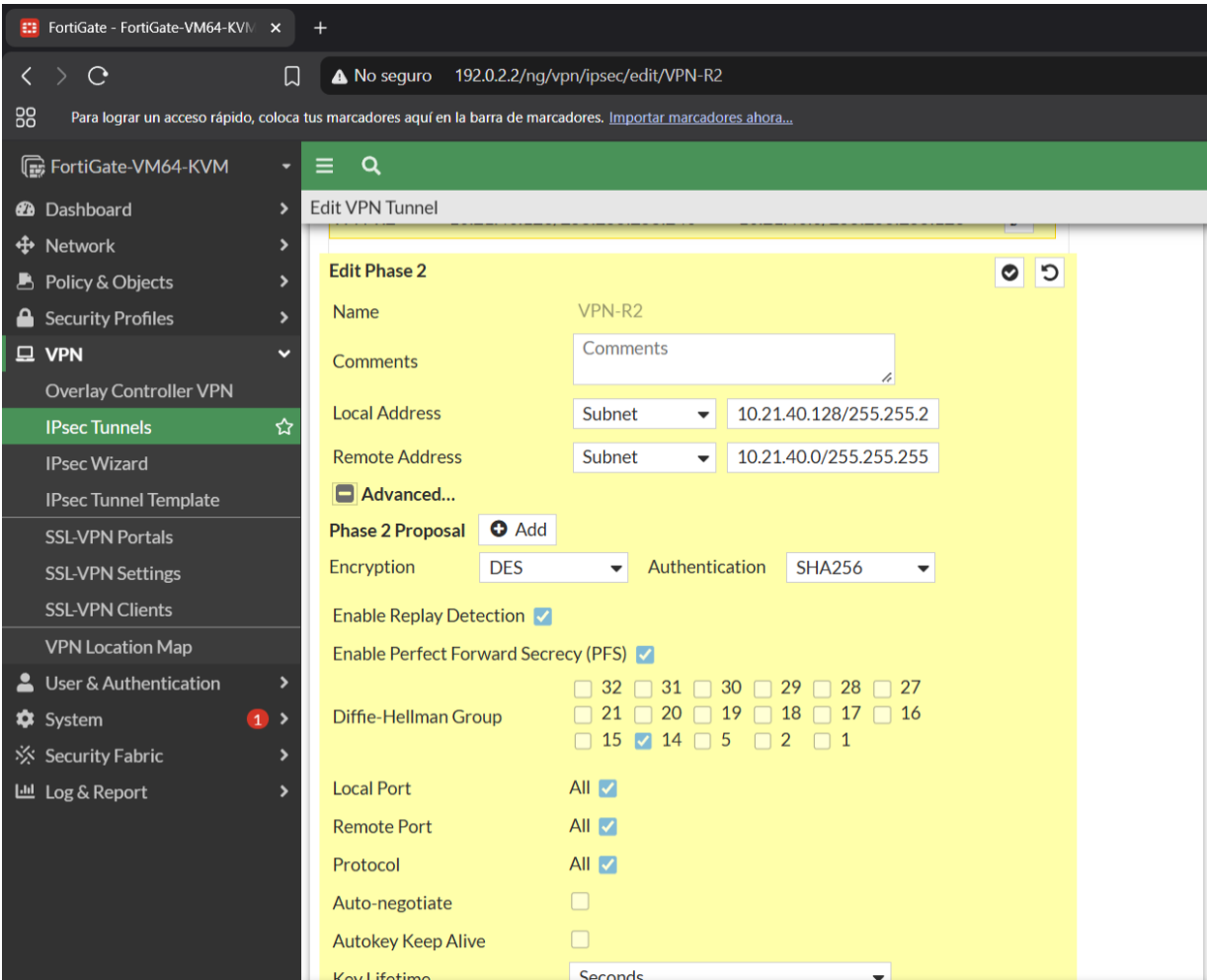
Diffie-Hellman Group						
☐ 32	☐ 31	☐ 30	☐ 29	☐ 28	☐ 27	
☐ 21	☐ 20	☐ 19	☐ 18	☐ 17	☐ 16	
☐ 15	☒ 14	☐ 5	☐ 2	☐ 1		

Key Lifetime (seconds): 86400

Nota: DES se utilizó por compatibilidad con la imagen del appliance disponible en el laboratorio. En un entorno de producción se recomienda utilizar algoritmos modernos como AES.

FortiGate - Phase 2 Proposal

La Phase 2 utiliza DES/SHA256, Replay Detection, Perfect Forward Secrecy (PFS) y Diffie-Hellman Group 14. Los puertos y protocolos se mantienen en All para permitir el tráfico definido por los selectores IPsec.



FortiGate - NAT para salida del servidor

La política SERVER-to-INTERNET permite que RED-SERVIDOR salga por port1 utilizando NAT con la dirección de la interfaz de salida. Los contadores de la política muestran sesiones y tráfico procesado, validando su uso.

The screenshot displays the FortiGate web interface for editing a firewall policy named 'SERVER-to-INTERNET'. The interface is in Spanish and shows the following configuration details:

- Name:** SERVER-to-INTERNET
- Incoming Interface:** port2
- Outgoing Interface:** port1
- Source:** RED-SERVIDOR
- Destination:** all
- Schedule:** always
- Service:** ALL
- Action:** ACCEPT (checked), DENY (unchecked)
- Inspection Mode:** Flow-based (selected), Proxy-based (unselected)
- Firewall / Network Options:**
 - NAT: ☒ (checked)
 - IP Pool Configuration: Use Outgoing Interface Address (selected), Use Dynamic IP Pool (unselected)
 - Preserve Source Port: ☐ (unchecked)
 - Protocol Options: PROT default
- Security Profiles:**
 - AntiVirus: ☐ (unchecked)

On the right side, the **Statistics (since last reset)** section shows the following data:

Statistic	Value
ID	3
Last used	10 second(s) ago
First used	49 minute(s) ago
Active sessions	21
Hit count	596
Total bytes	122.78 kB
Current bandwidth	0 bps

Below the statistics, there is a **Last 7 Days Bytes** chart showing traffic volume over time. The chart includes a legend for SPU and Software, with a bar for Software visible on Oct 02.

The interface also includes a sidebar with navigation options like Dashboard, Network, Policy & Objects, and a bottom status bar showing 'FORTINET v7.0.9'.

Conclusión

Las evidencias documentan la configuración realizada desde la GUI del FortiGate: interfaces WAN y LAN, rutas estáticas, objetos de red, políticas de firewall, NAT y el túnel IPsec Site-to-Site. El tráfico entre 10.21.40.0/25 y 10.21.40.128/28 utiliza la VPN sin NAT, mientras que la salida del servidor hacia redes externas utiliza una política independiente con NAT habilitado.

También se documentan los estados Up e Inactive de VPN-R2. La prueba de deshabilitación del túnel permite demostrar que la comunicación entre el usuario Windows y el servidor Ubuntu depende del enlace VPN.